

Each site has very different performance characteristics. Some spend a lot of time executing JavaScript, others spend a lot of time in marshaling, and some spend significant time in layout and rendering. To make a browser that is all-around fast, you have to understand how websites are built. For Internet Explorer 9, the development patterns used to build more than 7,000 real-world websites were studied, paying special attention to the top 1,000 websites worldwide. The analysis covered more than 50 dimensions to build a deep understanding of the patterns of those sites and took into account a range of performance characteristics including: time spent in different subsystems, use of common development frameworks like jQuery and DOJO, what analytics sites were using, how many script files sites were using, source code size, and also which features were being used.

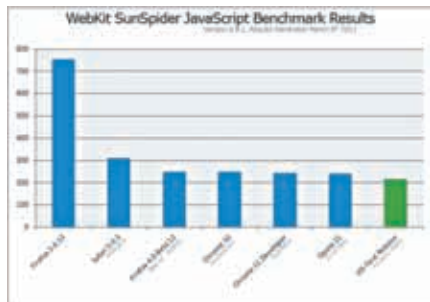
Hardware-accelerated graphics

Today, when you browse the web, you use about 10 percent of the power of your PC. Internet Explorer 9 unlocks the other 90 percent. Many browsers do this by taking a lowest common denominator approach to performance and minimizing PC capability. Internet Explorer 9 takes full advantage of the operating system and hardware to fully accelerate rendering of webpages. As an example, we use the graphics processing unit (GPU) when dealing with text, graphics, and new HTML5 technologies like Video, SVG, and Canvas.

Over the last 10 years, the computing power of the GPU has grown exponentially and today's GPUs can achieve more than one teraflop of computation. When you compare that to CPUs which have barely achieved 100 gigaflops of computation, you can see how much computation power can be unlocked through hardware.

Chakra, a brand new JavaScript engine.

Based on data, we know that most PCs have more than 2 CPU processor cores and Chakra, the new JavaScript engine in Internet Explorer 9, is optimized to take advantage of those multiple cores by compiling the JavaScript into highly



WebKit SunSpider JavaScript Benchmark